

	Document Type:	FORM ISO 9001:2015	Document Code	RES- F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
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(1) Research Program/Project/Study Title: CapSU Ginger Processing Technology: An assessment of its Production Benefits	
Author's Name:	Gerard Lee L. Atienza
Academic Rank:	Asso.Prof. V
Co-Authors (with academic rank):	Charina C. Gloria (Asso.Prof. V), Monalyn L. Oloroso (Asso.Prof. V), Marilyn L. Bernabe (Asso.Prof. I) & Arlyn C. Olmo (Prof.III)
Campus/Satellite College/Address:	Burias, Mambusao, Capiz
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Bayuyan, Pres. Roxas, Capiz
Province:	Capiz
Region:	VI
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)

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(7) Sector/Commodity	(8) Discipline/Field of Specialization
(9) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) This study was conducted to find out the production benefits of CAPSU ginger technology in terms of efficiency, quality of products and profitability. It employed mixed-method research design and data were gathered using a researcher-made questionnaire and semi-structured interview schedule. Face-to-face interview technique was done in gathering the qualitative data. Data were analyzed using descriptive statistics such as mean, frequency and percentage. Results of the study revealed that most of the respondents were senior citizens, married, have number of children of 3 and below, and completed high school, majority were housewives, and earned an income below Php5,000.00. Aside from being members of Bayuyan Orange Ladies Cooperative, respondents are also members of different organizations such as the Farmers' Association and Philippine Coconut Association. As to trainings attended, most of the respondents attended training on ginger tea making. With regard to technology efficiency and quality of ginger tea produced, respondents indicated that the machine is truly faster as compared to manual processing. Lastly, problems encountered by the adoptors are absence of independent electrification, low capital for ginger tea production, and non-spacious processing facility.	
(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) <i>Ginger tea production, efficiency, quality, and impact</i>	
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Ginger is widely cultivated as an ingredient in recipes and as a spice. It can be used raw, dried, juice or in powdered form. In 2020, ginger has gained considerable attention especially for its use in the treatment of inflammatory conditions. Ginger is a crop that grows best in the Philippines. In fact, it is so versatile that it can be grown almost any part of the country. But in spite of its suitability to Philippine conditions, ginger is still a “neglected” crop. Few farmers plant it and their produce is not enough to meet domestic consumption (Tacio, 2009). According to Food and Agriculture Organization of United Nations (2019), in the Harmonized National Research and Development Agenda 2017–2022 of the Department of Science and Technology, ginger is among the priority products under the functional food research and development component. The R&D thrusts were safety assessment of ginger for 2018 and establishment of health benefits and product development for 2019–2022. CAPSU Burias faculty specifically in the Engineering Department is on one way in attaining the goals of the Department of Science and Technology. They were able to developed a machine which could help in processing and producing ginger tea. Likewise, it is with their great desire to help the cooperative in Capiz by increasing the production of their ginger powdered tea. The researchers believed that CAPSU ginger processing technology would contribute significantly boost their production capacities. On this premise, this study was conducted.	

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Statement of the Problem

The objective of the study was to find out the socio-economic impact of ginger processing technology to the adoptors. Specifically, this investigation sought answers to the following questions:

1. What are the socio-economic characteristics of the local adoptors in terms of age, marital status, number of children, highest educational attainment, occupation, household monthly income, membership in organization, and training attended?
2. What are the production benefits of using the technology as compared to their manual processing in terms of efficiency, quality of their products, and production/profitability?
3. What are the problems encountered by the adoptors related to ginger technology?

(11) Materials and Methods or Methodology (This includes the materials, methods,duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.

This study was conducted at Bayuyan, Pres. Roxas, Capiz from November 2019 to November 2022. The participants of the study were the adoptor of the ginger processing equipment or the members of the Orange Ladies Cooperative. The data were gathered using a research-made questionnaire and the researchers conducted semi-structured interview schedule applying face to face interview technique. To avoid subjectivity in the gathering of data, assistance of the Municipal Agriculturist Office of the Municipality of Pres. Roxas was sought. Data were analyzed using descriptive statistics such as mean, frequency and percentage.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)
Beneficiaries Profile

Table 1 shows the distribution of the technology adoptor-respondents according to their profile.

Age

As shown in the table, one-third (33.3%) of the respondents were senior citizens (60 yrs old and above). This was followed by the respondents with age category of 40-49 years. Same percentage (19.4%) of respondents belonged to the age bracket of 30-39 years and 50-59 years, respectively. The lowest age group was shown among below 30 years respondents (5.6%).

Marital Status

More than half (52.8%) of the respondents were married, less than a third (30.6%) were widowed/separated, and 16.7 percent were single.

Number of Children

In terms of the number of children, majority (63.9%) of them had 3 children and below, more than one-fourth (27.8%) had 4-6 children and a little percentage (8.3%) had 7-9 children.

Highest Educational Attainment

The same table shows that most (30.6%) of the respondents were high school graduates, one-fourth (25%) of them were high school level, while 16.7 percent were college level and 11.1 percent were college graduates. The same percentage (8.3%) was shown by respondents who earned elementary level and elementary graduate, respectively.

Occupation

Data show that majority (55.6%) of the respondents were housewives, one-third (33.3%) were self-employed, and few percentages (8.3% and 2.8%) were government and private employees, respectively.

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Monthly Income

As to their monthly income, majority of the respondents had an income below Php5000, while some of them (11.1%) had an income of Php10,000 and above, and only 2.8 percent had an income of Php5000 to Php9999.

Membership in Organization

Aside from the respondents' membership in Bayuyan Orange Ladies Cooperative, they are also members of different organizations such as the Farmers' Association (33.3%), Phil. Coconut Association (11.1%), Kalipi (16.7%), and RIC (5.6%).

Training Attended

Of the 36 respondents who were interviewed, majority or 69.4% attended training on ginger tea making for 8 hours sponsored by the LGU, DOLE and CAPSU.

Production Benefits of Using the Technology

Efficiency

When respondents were asked about the benefits of using the technology compared to manual processing in terms of *efficiency*, respondent 1 claimed that they do not exert much effort and it is not laborious. Respondent 2 added that it became faster compared to manual processing, while respondent 3 claimed that it shortened their processing time, more products are produced, and less labor. Respondent 4 said it is no longer tiring to process the ginger tea. On the whole, the technology in terms of efficiency is beneficial as perceived by the respondents.

Quality of Products

In terms of the *quality of products*, respondent 1 claimed that it has better quality in terms of texture and color. Respondent 2 claimed that it has good quality, *wala na bilog-bilog ang powder, and okay gid ang texture kag puede na mabaliya*. Respondent 3 said that the product is of better quality, it has fine texture, crystallized and rightly cooked. Respondent 4 also *claimed na mas mapino, mas crystallized and sakto ang pagkaluto*. With all the responses mentioned by the respondents, it is say to mention that the technology used by the cooperative is able to generate a better quality of ginger tea.

Production/Profitability

In terms of *production capacity*, respondents claimed that their produced became doubled. In 2019, their produce was only 20 kilograms, while in 2022 they were able to produce an estimated of 50 kilograms of ginger tea per year.

As regards to Return on Investment, respondents mentioned that their sales from producing ginger tea is spent for the improvement of their processing facility.

In terms of market reach, the respondents mentioned that their ginger products are no longer sold to external buyer. Instead, their produce is sold to their cooperative as a the main ingredient in the making of ginger tea. With this result, it is safe to imply that with the presence of the technology in their cooperative has become advantageous and easier for them to market and to earn money from their produce. Further, the ginger tea they produce is already being displayed in the University ATBI and is sold to local consumers.

Problems Encountered by Adoptors

Respondents claimed they did not encounter problems concerning technology except for faulty dynamo. They said that if the machine needs to be fixed, they inform CAPSU to troubleshoot the problem to make it functional.

Further, they likewise experienced problems among members of the organization. Other members complained about the participation of other members in cooking. They suggested that they should have work scheduling and shifting of roles.

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Electrification is also one of the reasons that hampers the cooperative's operation. They do not have an independent connection going to their facility. They suggested that electric meter and electrical connection must be installed. Likewise, the processing facility is not fully structured, and it is not so convenient due to the limited space. They mentioned that the said facility needs to be enhanced. Lastly, the organization lacks of budget as additional capital for continued production of ginger tea.

Table 1. Profile of the respondents.

VARIABLES	FREQUENCY	PERCENT
Age		
60 years old and above	12	33.3
50-69	7	19.4
40-49	8	22.2
30-39	7	19.4
Below 30 years old	2	5.6
Marital Status		
Single	6	16.7
Married	19	52.8
Widowed/Separated	11	30.6
Number of Children		
3 and below	23	63.9
4-6	10	27.8
7-9	3	8.3
Highest Educational Attainment		
College Graduate	4	11.1
College Level	6	16.7
High School Graduate	11	30.6
High School Level	9	25.0
Elementary Graduate	3	8.3
Elementary level	3	8.3
Occupation		
Government employee	3	8.3
Private employee	1	2.8
Self-employed	12	33.3
Housewife	20	55.6
Monthly Income		
Php10,000.00 and above	4	11.1
Php5,000.00 - Php9,999.00	1	2.8
Below 5,000.00	31	86.1

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Table 2. Membership in organization.

NAME OF ORGANIZATION	FREQUENCY	PERCENT
Bayuyan Orange Ladies Cooperative	36	100.0
Farmers Association	12	33.3
Kalipi Association	6	16.7
Philippine Coconut Association	4	11.1
RIC	2	5.6

Table 3. Training Attended by the respondents.

Training Attended	FREQUENCY	PERCENT
Ginger tea making	25	69.44

Table 4. Name of Training Attended by the respondents.

Name of Training	No. of Hours	Sponsoring Agency
Ginger tea making	8	LGU, DOLE, CAPSU

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

1. The machine was found beneficial by the adoptors in terms of efficiency as compared to manual processing and in terms of quality as the technology is able to produce better quality of ginger tea.
2. Adoptors encountered problems in terms of electrification, processing role, not spacious processing facility, and capital for the production of ginger tea.

Recommendations

1. CAPSU with the help of the Municipal Agriculturist Office of the town of Pres. Roxas, should continue monitoring and should extend help and assistance for the installation of electrical lines and meter to improve the processing facility.
2. Brgy. Bayuyan should allot enough budget for the expansion and improvement of the processing facility to make the workplace more comfortable and ideal.
3. Orange Ladies' Cooperative should outsource budget to increase their capital in the production of ginger tea by topping other benevolent persons and agencies.

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14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Food and Agriculture Organization of the United Nations. (2019). Ginger Value Chain Study in Nueva Vizcaya, Philippines (GCP/RAS/296/JPN). Retrieved from <http://www.fao.org/3/ca5767en/ca5767en.pdf> on November 15, 2020.

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